

Final Design Review (FDR)

TerraProbe

A Down2Earth (D2E) Company

5/1/2025



School of Mechanical Engineering

Meet the Team



**Awadhoot
(Avie) Ghatge**



**Jacob
McKenrick**



**Lokesh
Sriram**



**Sankaran
Iyer**



**Christopher
Meyers**

CDR Recap

Reflecting on our ideation and design process



School of Mechanical Engineering

CDR Recap – Problem & Customer Requirements

Soil Testing Challenges: Costly, Slow, and Inaccurate

- **Soil Testing in Agricultural Industry**
 - Important!
 - Soil testing process is not optimal
 - Inaccurate, timely, expensive
- **Solution & Customer Requirements (Farmers)**
 - Easy to operate
 - Accuracy in data
 - Portable
 - Time Saving



CDR Recap – Competitive Landscape

Current Soil Sampling Solutions are expensive, complex, and outdated

		Portability	Cost	Sampling Depth	Operation Type	Durability	Real-Time Data
	AMS Soil Probes	Weight: ~1 to 3 lbs Length: ~33 inches	\$100-\$150	8"	Manual	Stainless Steel, requires minimal maintenance	No
	WintexAgro 1000 Automatic Soil Sampler	Weight: 105 lbs, Requires mount on Vehicle	\$8000	11.8"	Designed for vehicle sampling, automated sampling	Requires heavy maintenance due to bulky equipment	No
	Amity Technology Soil Sampler	Weight: 210 lbs Requires mount on vehicle	\$3000-\$6000	24 to 48" (Based on model)	Designed for vehicle mounting, automated sampling	Requires standard maintenance	Yes, partners with 3 rd party to provide (FARMQA)

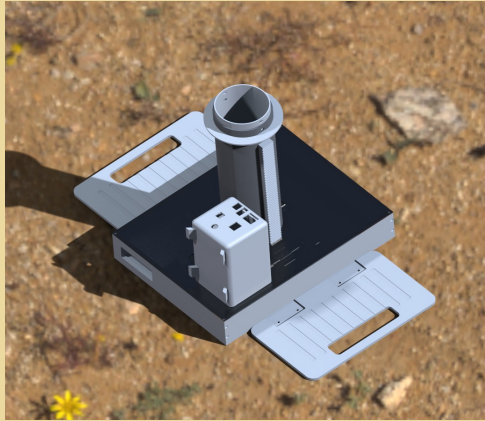
Market Gap for an Intermediate Solution

- **Cost:** High-end solutions cost \$5,000+, while cheaper alternatives lack precision and automation.
- **Operation:** Advanced models require skilled operation, while manual tools are slow and labor-intensive.
- **Insights:** Lab-based testing takes days, preventing real-time decision-making for farmers and researchers.

Recap of TerraProbe Products

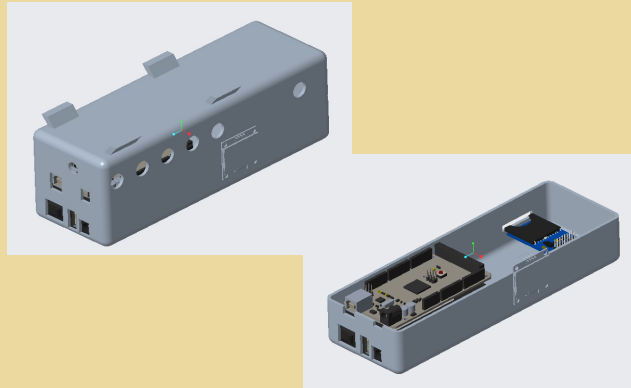
All-in-one solution encompassing burrowing, testing, and analytics

TerraProbe Soil Sampler



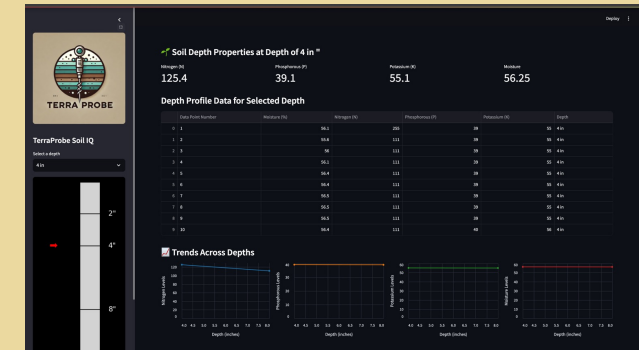
- ✓ Compact, **Portable** & Handheld Soil Sampling Device (9"x 12"x 17")
- ✓ Utilizes **Rack and Pinion Mechanism** for motorized power and downward movement (12" Deep)

TerraPal Testing Probe



- ✓ Sensor Module for **on-site analysis** capturing key parameters
 - **Nitrogen, Phosphorous, Potassium (NPK)**
 - **Moisture**
- ✓ Readings taken at **slits on the inner payload**

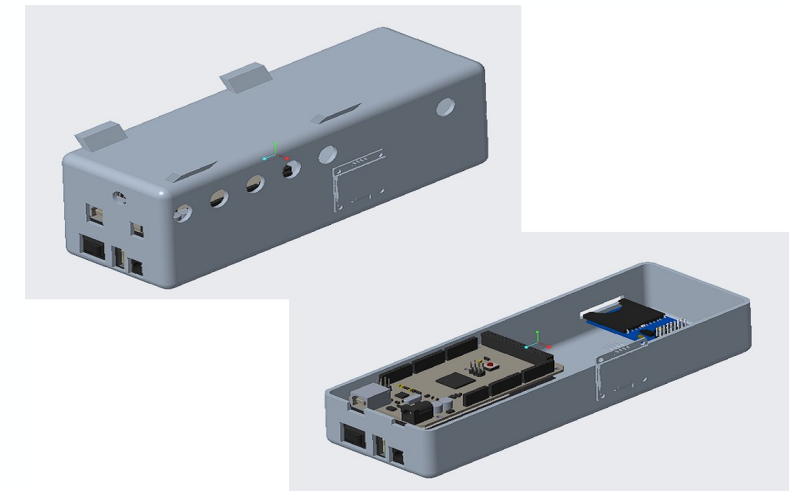
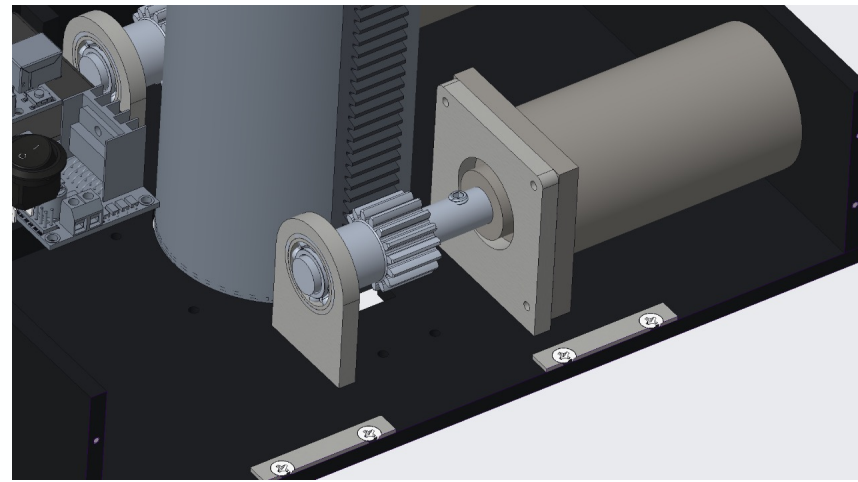
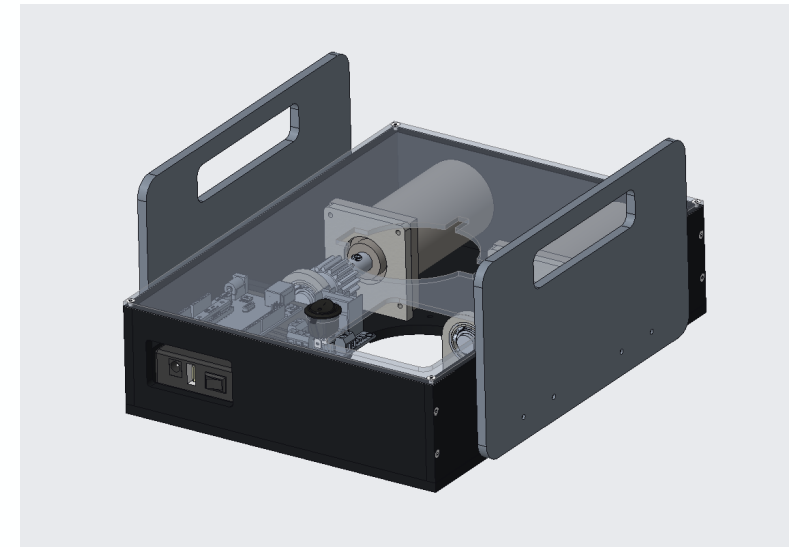
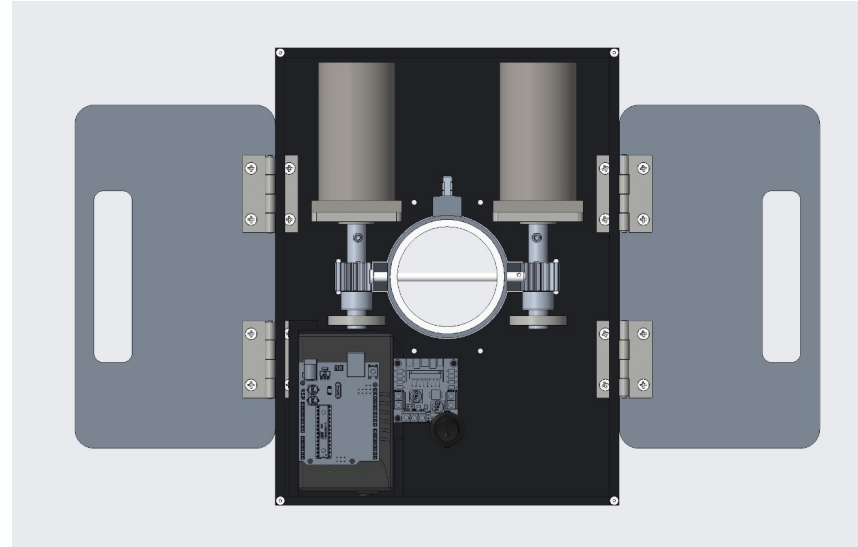
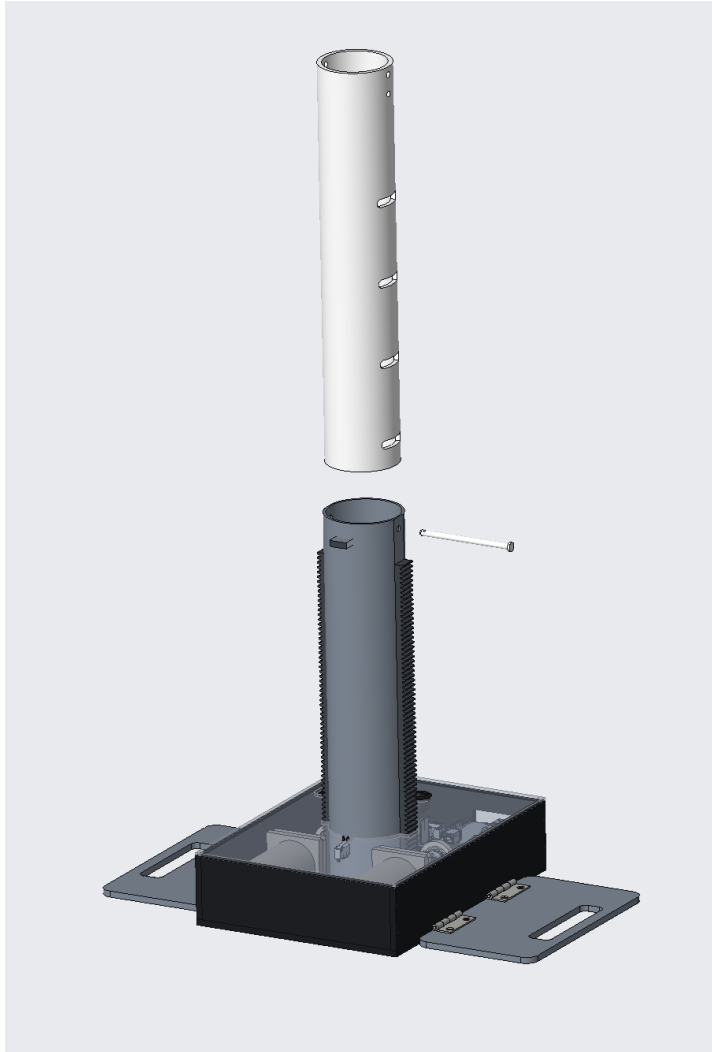
SoilIQ Dashboard



- ✓ Digital platform to **visualize NPK and moisture data** at various depth profiles
- ✓ Converts **SD card data from DAQ** into interpretable tables and charts

CDR Recap: TerraProbe's Sub-Assemblies

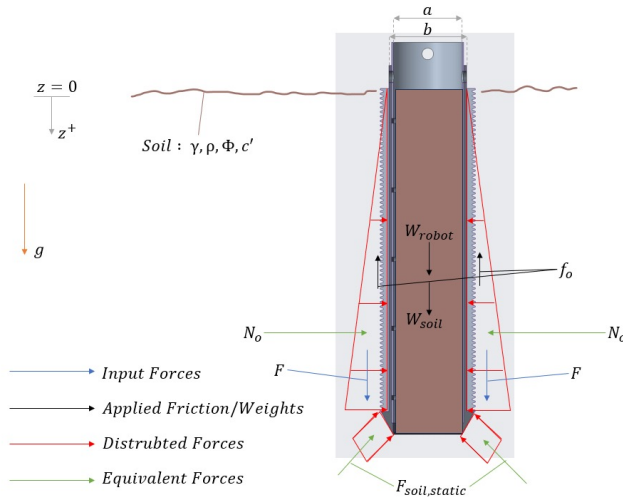
Payload for sampling, gear-shaft for power, and probe for real-time analysis.



CDR Recap: Design for Manufacturing

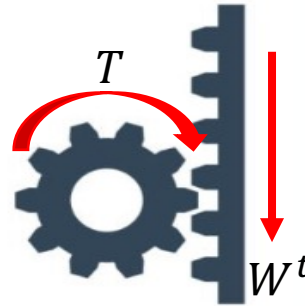
TerraProbe chose its parts based on in-depth analysis

Motor Selection



- The **Rankine Theory** : used to compute the lateral soil pressure on the sides of the guidance system
- **Motor Requirements**
 - 35 W and 27.5 RPM
- **Calculated Force**
 - 923 N

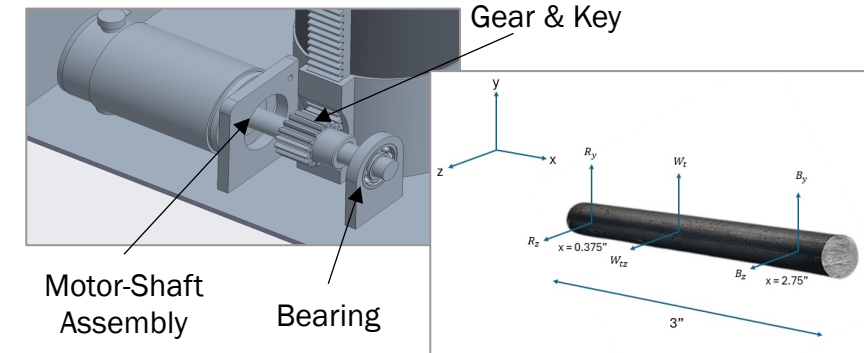
Gear Analysis



Pinion and Rack choices: (*All Choices follow AGMA Standards)

- **Wear:** $S_H = 1.93$
- **Bending:** $S_T = 3.49$
- Minimum # of Teeth – 16
- Pinion Diameter – 1 inch
- Rack/Pinion Pitch – 16 teeth/ inch
- Material : AISI 1144 Hardened Steel
- Parts from McMaster Carr

Shaft, Key, Bearing Design



Influence of Shaft, Key, & Bearings Design

Shaft (Diameter 0.5", Length 3" – AISI 4140 Steel)

- Connects the motor to the gear system
- Fatigue Factor of Safety: 4.4
- Yielding Factor of Safety: 8.8

Key (0.125" x 0.125" x 0.25" – Square Key, AISI 1045 Steel)

- Critical for tight fit with the gear
- Shear & Crushing FOS: 1.5 (AGMA Standard Compliant)
- Torque Transmission: 106 lbf-in

Bearings (McMaster-Carr 60455K291)

- Prevents failures due to excessive stress, misalignment, or wear
- Dynamic Load Capacity: 1140 lbf
- Max Speed: 25,000 RPM

Manufacturing & Assembly

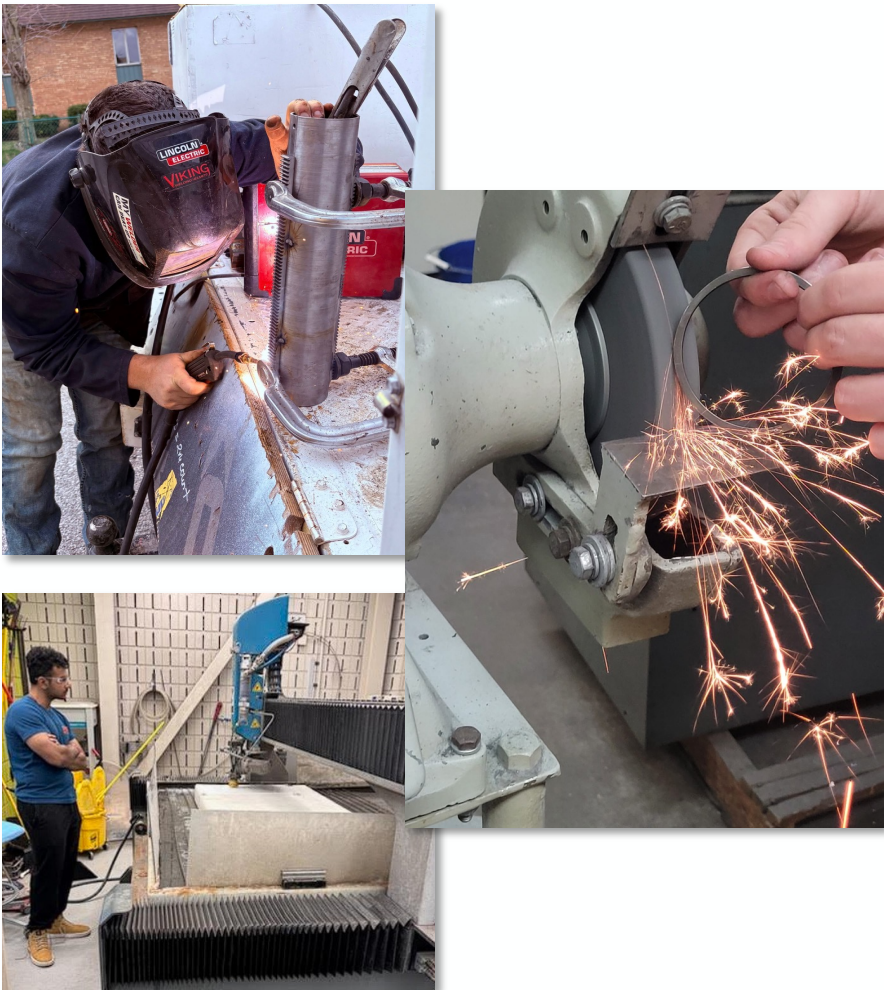
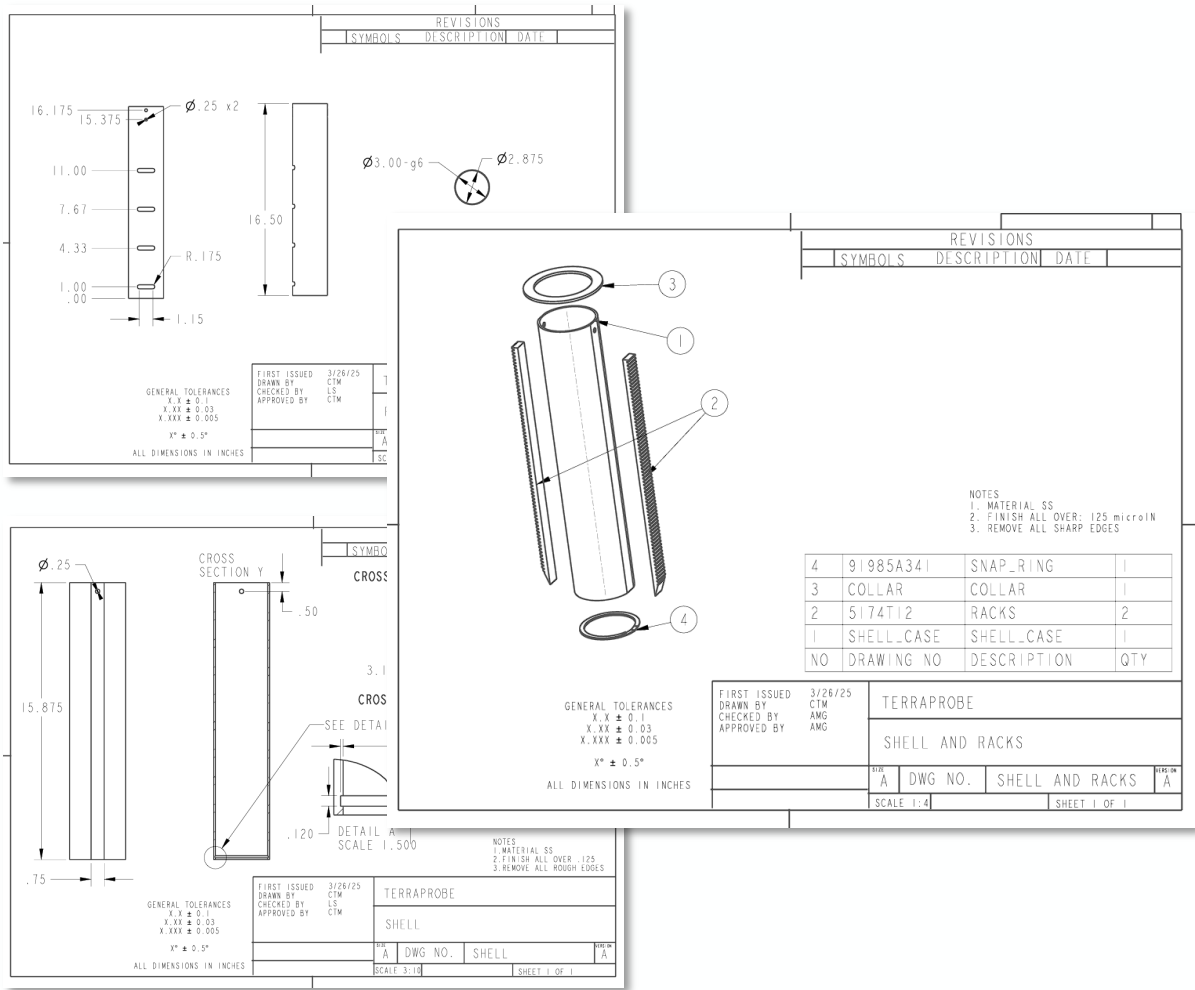
Reflecting on our ideation and design process



School of Mechanical Engineering

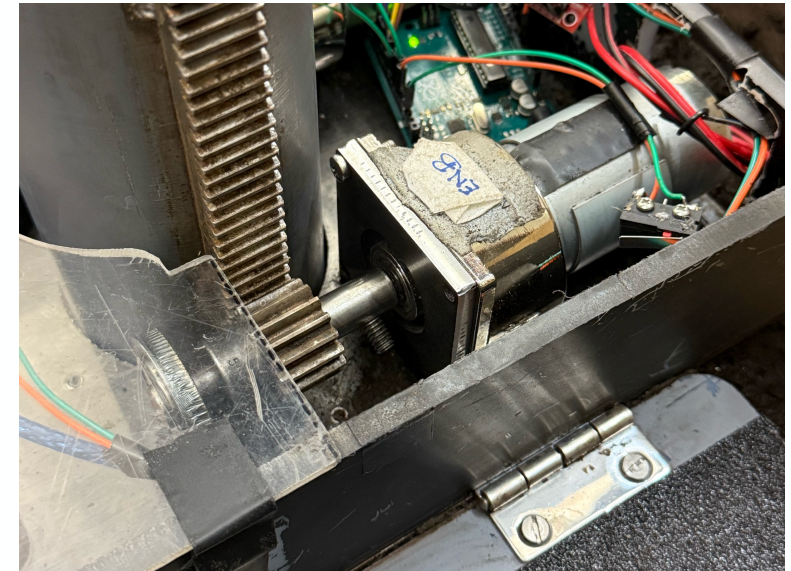
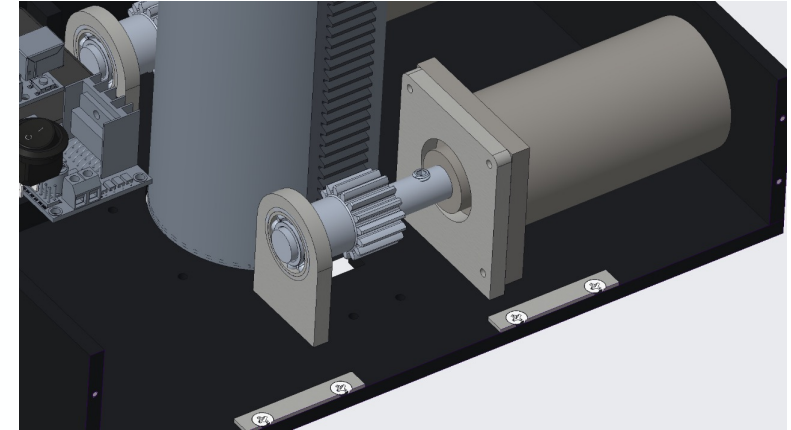
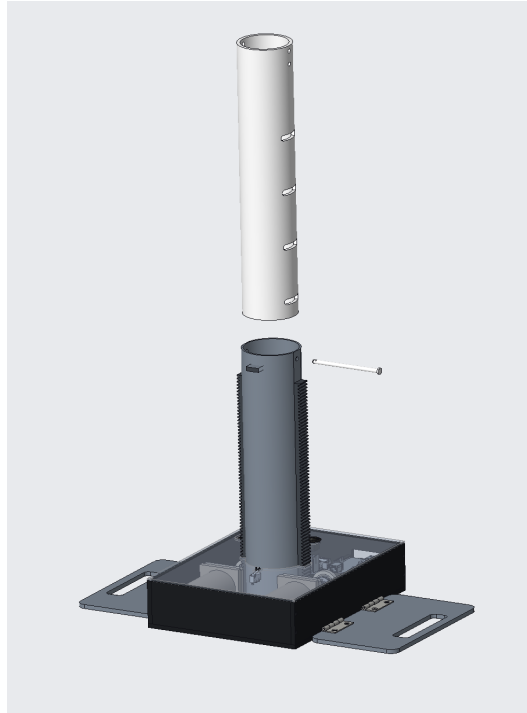
Manufacturing Drawings and Process Plans

TerraProbe was carefully crafted with manufacturing/welding drawings and detailed machining



TerraProbe - CAD vs. Production Prototype

Comparison of Soil Burrowing and Sampling System Design & Prototype



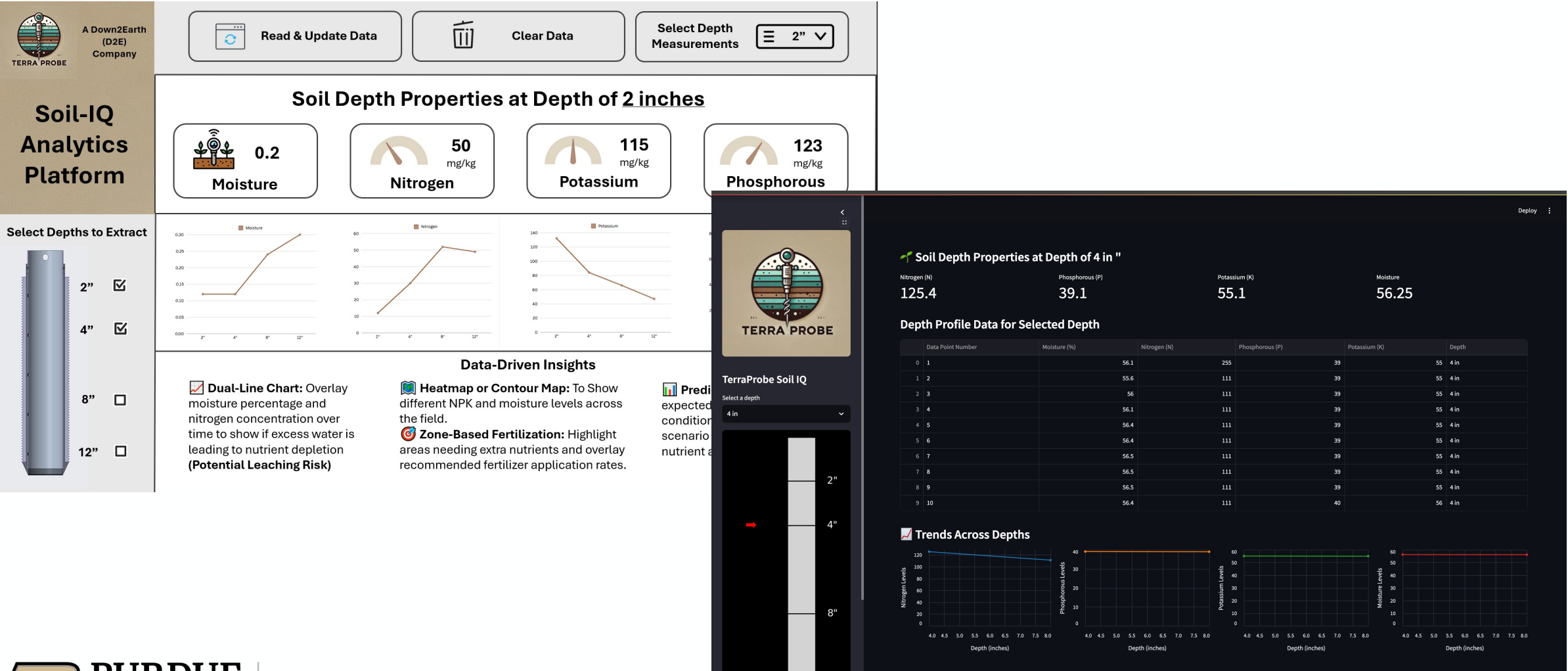
TerraPal Testing Probe - CAD vs. Production Prototype

Comparison of Soil Testing System CAD & Prototype



TerraSoillQ Dashboard - Design vs. Production Prototype

Comparison of SoillQ Dashboard System Design & Prototype



TerraProbe - Burrowing



PLACE DEVICE ON SOIL AND STAND ON FOOT PEDALS

TerraProbe - Extraction



TerraPal - Testing



TerraSoillQ - Dashboard

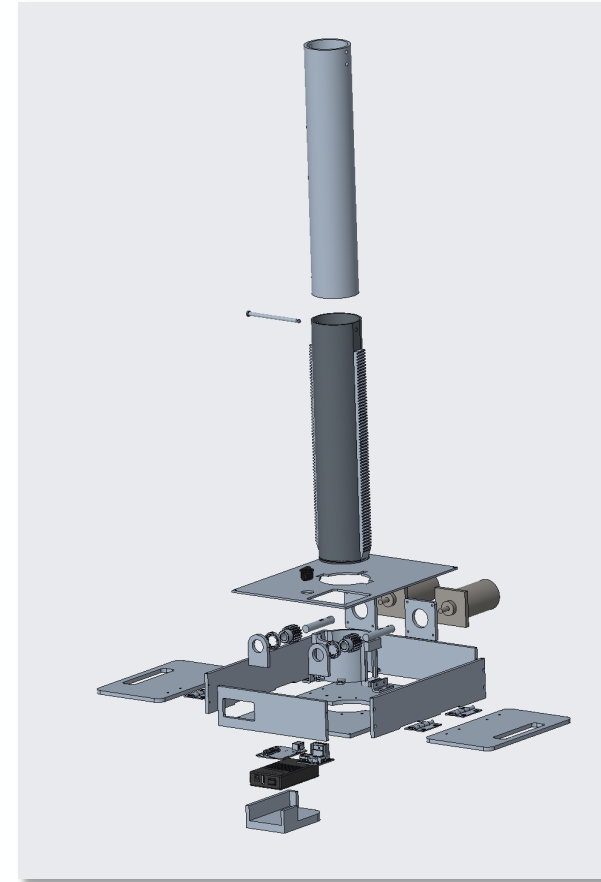
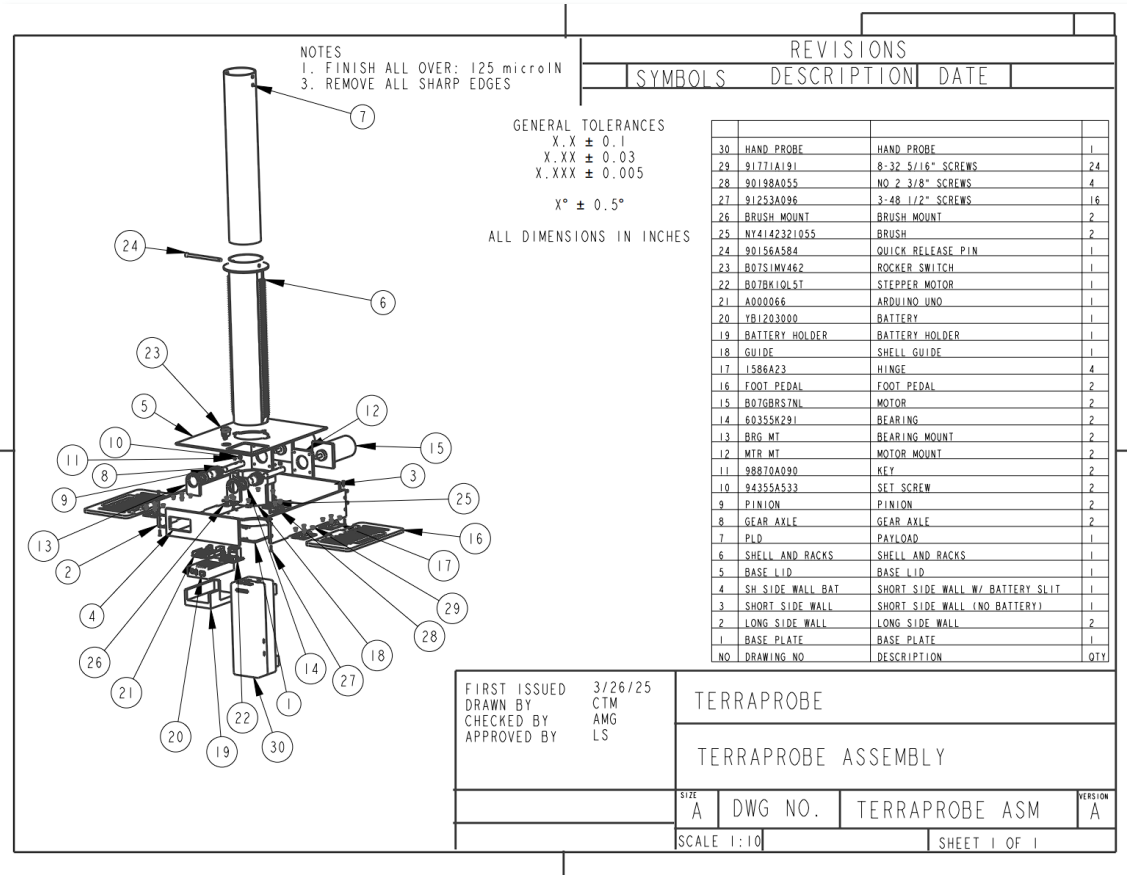
TERRAPROBE SOILIQ DASHBOARD

Validation Tests

Motors, TerraProbe, TerraPal, and TerraSoilIQ were all tested and analyzed for their failure modes

CAD Refinements: Key Changes from CDR

The TerraProbe team took manufacturing constraints into account to update CAD models and final design



- Base plate welds substituted for bolts
- Added lock-nuts to foot pedal screws
- Removed shell guide
- Trimmed metal brushes

Functional Requirements – Targets vs. Actual Performance

TerraProbe partially meets the functional requirements outlined

Requirement	Target	Actual
Portability	< 25 kg, compact	11.8 kg, 9" x 12" x 16.5" - Compact
Motorized Operation	Run independently without user effort	Has Motorized Operation but requires user supervision and careful assembly
Auger/Core Sampling Mechanism	Sample down to 12 inches	Samples down to 6.5 inches
Depth Accessibility	Distinct samples from 0–2 in, 2–4 in, 4–8 in, 8–12 in	Distinct samples from 0–2 in, 2–4 in
Soil Type & Compactness	Device burrows through different amount of soil compactness	Easily burrows through cohesive soil but not dry soil
Motor Control	Provide a minimum of 30 RPM force	Provides ~22.5 RPM (Lower than expected) 120 lbf MAX Force and 90 lbf Steady State
NPK Data Capture	Values for Nitrogen, Phosphorus, and Potassium	Functions as intended and captures moisture as well
Dashboard ML Model Accuracy	>= 85% Model Performance Accuracy	92.2% Model Accuracy
Cost Efficiency	Prototype development cost from \$600-1000, with sell price <\$1500	Prototype development cost ~\$980

Validation Tests – Force Test

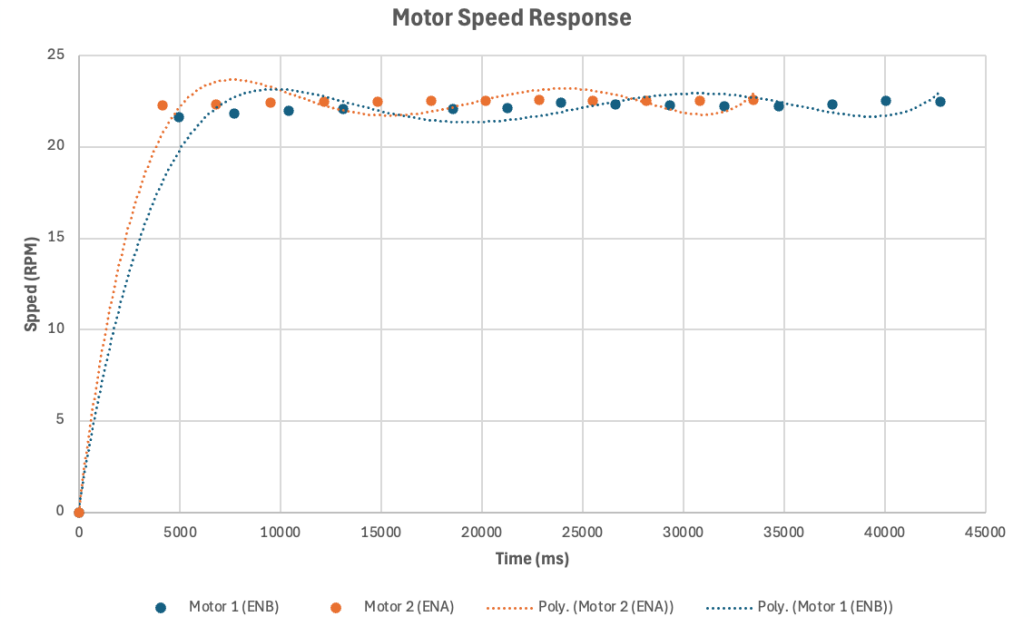
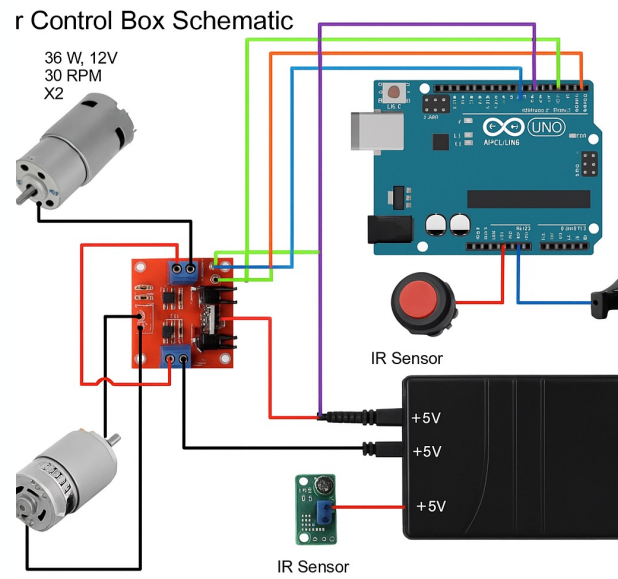
Future scope will require higher powered motors for faster and optimum user experience



- ✓ **Force Test**
 - Placed a scale on top of the shell and payload to examine force to burrow down
 - Pushed down to obtain maximum force value that can be applied using hands
- ✓ **Implications**
 - Force required to push down using hands is 120-130 lbf
 - Motors provide a max force of 120 lbf with 90 lbf steady state
 - Collected around 6-7 inches in both manual and actual test
- ✓ **Takeaways**
 - Perform motor tests to identify RPM and Power impact on motor

Validation Tests – Motor Performance

Future scope will require higher powered motors for faster and optimum user experience



✓

Motor Design & Sizing

- Selected a minimum of **35W, 12V motor at 30 RPM**
- Required force calculated as 923.4 N for stiff cohesive soil

✓

Motor Speed Test

- Connected IR sensor to measure the pulses per revolution of the motor

✓

Motors are underperforming

- Performed IR Sensor Test to examine motor speed response data over time
- Motors are running at MAX speed
- Motors steady state speed: ~22.5 RPM**
- Motor Rise Time: ~5 seconds**

✓

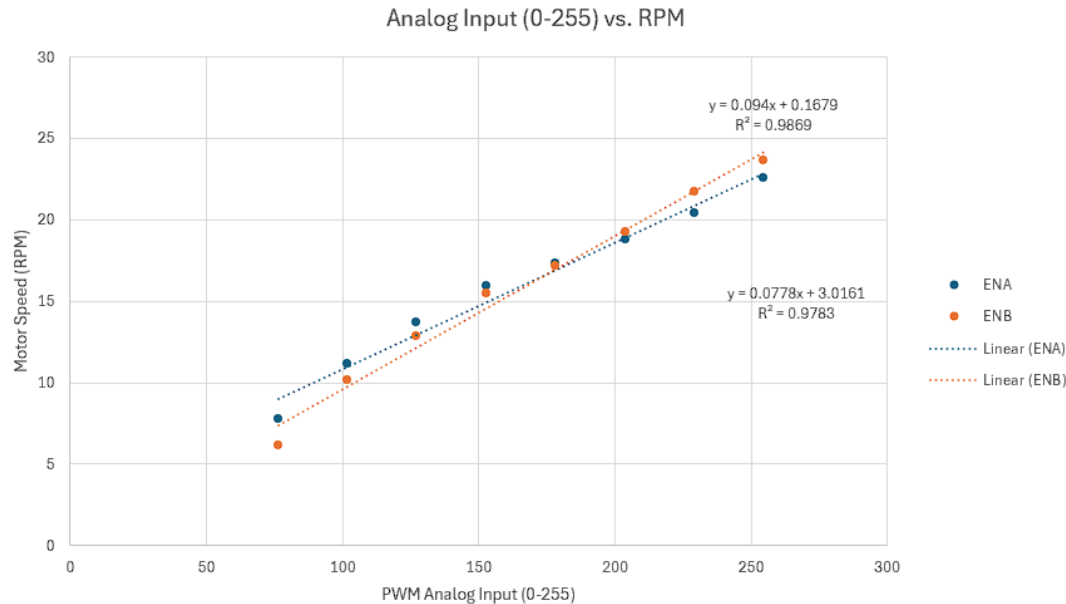
Implications

- Longer time to burrow into the soil
- 50 RPM motors necessary

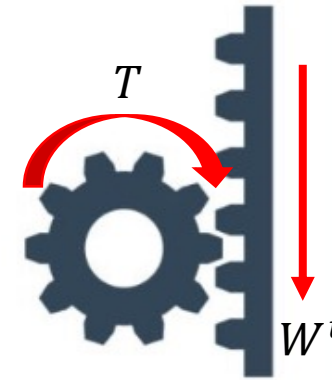
Validation Tests – Motor Calibration & Depth Testing

PWM Motor Output and Depth Performance testing

- ✓ **Motor Calibration – PWM Input vs. RPM Output**
 - Baseline to understand how motor speed affects soil penetration
- ✓ **Linear Motor Performance**
 - **Goal:** Run both motors at the same speed to avoid slip/tilt
- ✓ **Stiction**
 - Motors begin rotating only above ~30% input (≈ 76.5 PWM). Below this threshold, torque is insufficient to overcome static friction



RPM vs. Torque



- ✓ **Low RPM $\rightarrow$ Higher Torque**
 - Applicable for initial soil penetration due to requirement of force)
- ✓ **High RPM $\rightarrow$ Lower Torque**
 - Suitable for faster motion but offers less pushing force – not ideal in compact soil

RPM	ENA PWM	ENB PWM	Depth (in)
22.8	255	255	3.5
22	244	232	2.75
18	192	190	~1" (Did not run)
14	141	147	Did not run

- ✓ **RPM vs. Depth**
 - Lower RPMs did NOT result in any measurable burrowing, despite motors running
- ✓ **Why Low RPMs Failed?**
 - PWM controls power, not just voltage
 - Longer PWM cycles reduced the average current delivered

Validation Tests – Soil Compactness vs. Depth Performance

TerraProbe tests how well it can perform on wet vs. dry soil



**Base Material: Dry Garden Soil
(5174.8 cubic inches)**

Test Procedure

- 1) Base material: Dry garden soil (~5,000 in³)
- 2) Incrementally added 128 oz of water per test (~5% volume each time)
- 3) Measured depth burrowed (inches) after each addition
- 4) Simulates real-world changes in soil density and compactness

Soil Condition

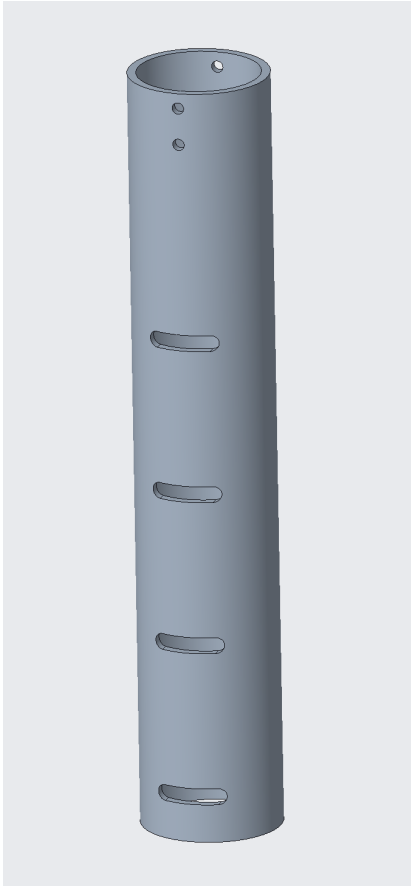
- ✓ Powdered
- ✓ Dry & Loose
- ✓ Lightly Moist (Slightly Cohesive)
- ✓ Moist (More Cohesive)
- ✓ Moderately Dense (Slightly sticky)
- ✓ Dense (Sticky)

Water Added (oz)	Approx. Condition	Depth Burrowed (in)
0	Powdered	1.21
128	Dry & Loose	1.75
256	Lightly Moist	2
320	Lightly Moist	3.93
384	Moderately Moist	5.56
512	Dense	6.72

- ✓ **Initial High Resistance**
 - Dry soil offers high resistance due to lack of cohesion.
 - As moisture increases, soil cohesion improves, making it easier for the probe to burrow.
- ✓ **Higher Moisture Levels, Greater Depth Sampling**
 - At high moisture levels (384–512 oz), soil becomes sticky and compact, which aids in deeper penetration by maintaining contact and reducing kickback.

Validation Tests – NPK & Moisture Sensor Precision

TerraProbe tests how well the testing probe precision performs



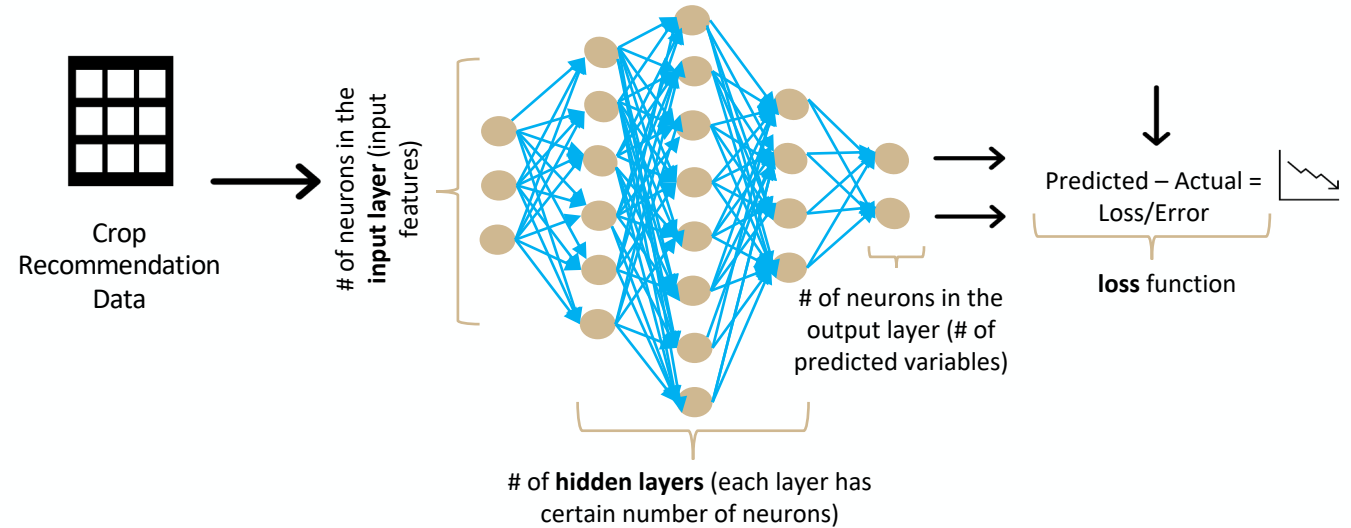
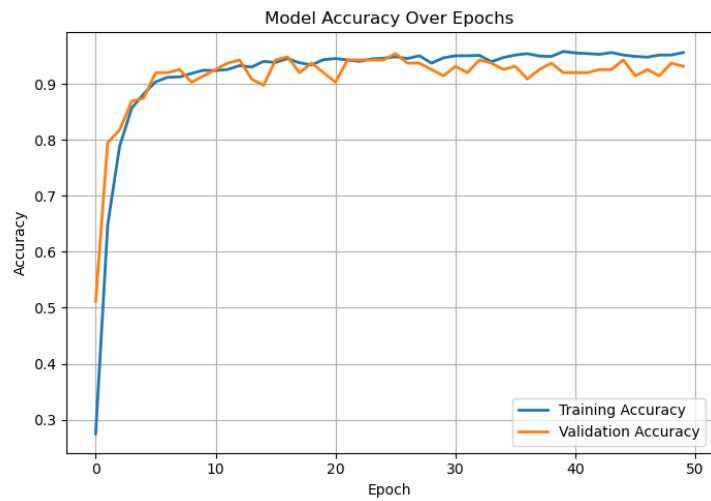
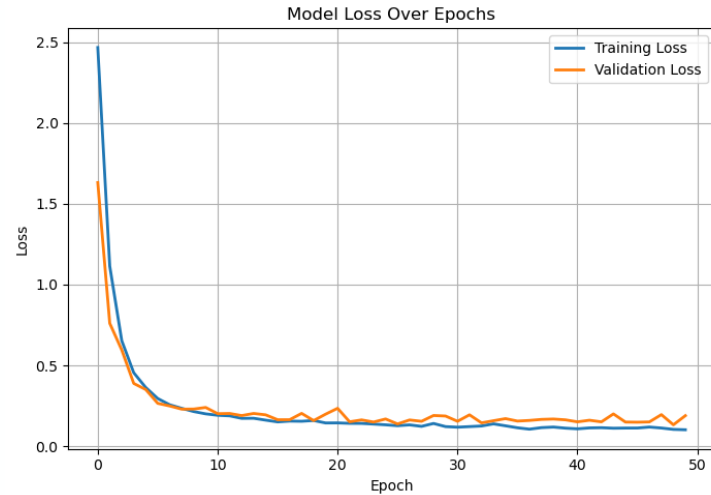
*Separate testing on Soil Cup

Trial Number	Nitrogen [mg/kg]	Potassium [mg/kg]	Phosphorous [mg/kg]	Moisture (%)
1	8	3	4	~67%
2	8	2.5	4	~69%
3	8	2	4	~69%
4	8	3	4	~66%
Average	8	2.625	4	~67.75%

- ✓ Conducted **4 repeated trials** on the same soil sample to assess precision of TerraProbe readings
 - Measured Nitrogen (N), Phosphorus (P), Potassium (K) levels and Moisture content
- ✓ Observed consistent Nitrogen (8 mg/kg) and Phosphorus (4 mg/kg) across all trials
 - Potassium showed minor variation (range: 2–3 mg/kg)
 - Moisture readings stayed within a tight range (~66–69%), indicating stable sensor output

Validation Tests – Neural Network Performance

Neural Network based crop recommendation results are trustable and generalizable



- ✓ **Train-Test-Split (80% Train, 20% Test)**
 - Split crop recommendation data into different segments
 - Tested model on known data to identify accuracy
 - Model Accuracy: 92% (Greater than 85% Industry Standard)
- ✓ **Low to Negligible Model Loss**
 - Loss indicates the difference between actual and predicted values
 - Low loss indicates that model performance is accurate and can be generalizable
- ✓ **Improvement and Limitations**
 - Data is based on aggregate India parameters (NPK, Temperature, Humidity) – More data from other countries could help improve model generalization

Improvements: Potential Failure Modes Simulation FMEA

Future scope will require higher powered motors for faster and optimum user experience

 Failure Modes	 Design Improvement
Soil is not compact enough to stick to payload	Added grip tape inside payload chamber to hold soil via friction
Motors not providing enough power	Plan is to replace motors with motors that maintain consistent power at lower RPM (voltage-controlled)
Motor direction unknown on startup	Programmed motors to default to forward motion (rack-down)
Wires interfering with gear/rack system	Mounted wires along sidewalls and added clips for better wire management
Foot Pedals breaking off due to excessive force	Used longer screws and attached bolts to the back-end
Top Lid made maintenance difficult	Redesigned lid by cutting into halves for quicker access to electronics (Future Plan: Add hinges to open and close lid anytime)
Shell Guide creates an interference with the rack	Removed shell rack from design as gears align smoothly with rack itself
Motor Performance deteriorates with lower battery	Batteries should always be ran fully charged, better and more powerful 12V batteries to be purchased for future iterations

Learnings & Improvement Opportunities

TerraProbe will take key insights from the testing of production prototype to improve design & manufacturing



Higher Torque Motors



Current motors don't provide **sufficient torque** for 12" depth



Battery life affects motor speed



As battery discharges, the internal source resistance increases, and less voltage appears across the terminals



Manufacturing Prototype

Larger base, cleaning protocol, and better cable management can be used for future iterations



Material Selection

Foot pedals are a failure point and steel could have been used rather than UHMW plastic



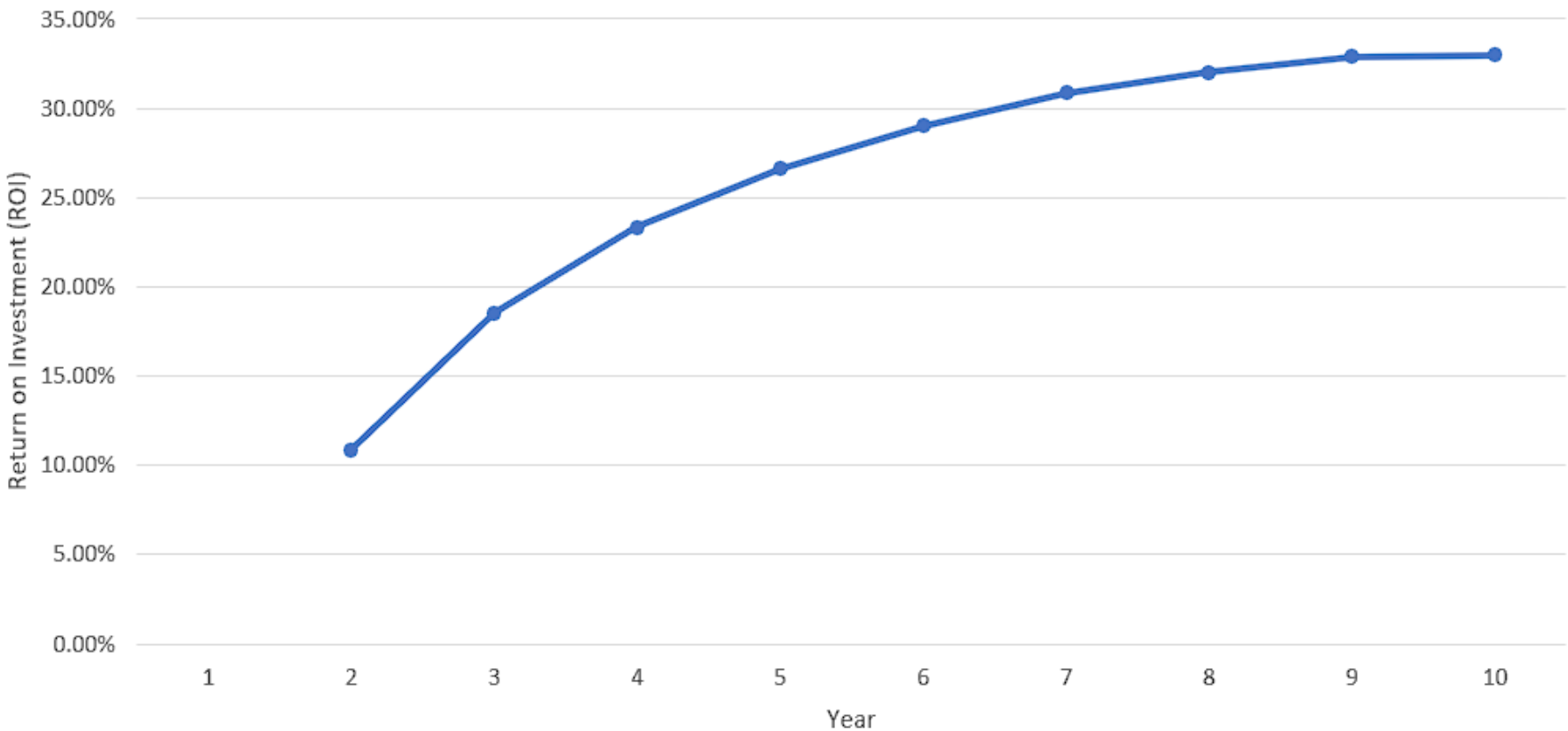
DAQ System

Better holding case (handle), smaller circuit and PCBs could replace majority of wiring for the system

Economic Analysis

TerraProbe presents strong scalability and projects to break even in ~3.5 years

Estimate Return on Investment by Year



Targeting 20% of Small/Medium Acreage Farms over 10 years (~175,000 units)

Production Cost
TerraProbe → \$862.77
Inner Tube → \$84.24
Margin & Sell Price - TerraProbe
25% GM → \$1100 MRP
Margin & Sell Price – Inner Tube
50% GM → \$130 MRP
10-Year Annual Revenue
\$50M
10-Year ROI
33%
3/31/23
⟨#⟩

Value Proposition & Path Forward

TerraProbe is on track for viable impact

Current State

☑ Minimum Viable Product (MVP) Achieved

🔧 Key Strengths: **Portability, Real-Time Data Acquisition, & Dashboard**

Path Forward

🚚 Motor Torque Limitations, Wiring Reliability, Sizing & Assembly

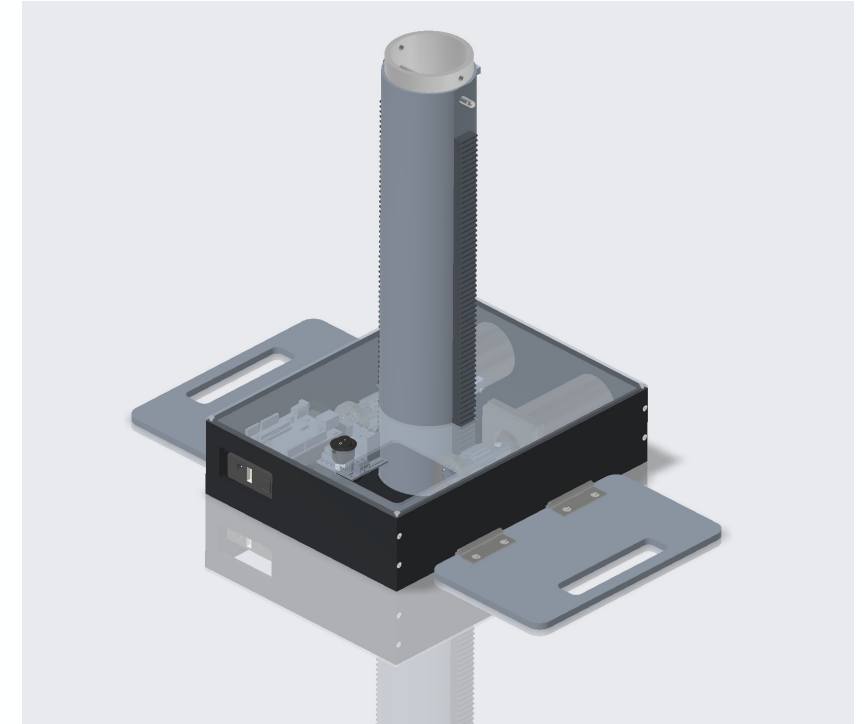
☢ Component Upgrades, Mfg. Refinements, Electronics Enhancements

Business Impact

🌉 Intermediate– Low-end usability and high-end features

🔥 Empower Small Farms – Data-driven decisions for better yields

💰 Grow to a \$50M Annual Revenue, Capturing 20% of the Market Share



Thank you

Any Questions?



School of Mechanical Engineering

References

Following are used in the development of this FDR

Crop Nutrition. (n.d.). *Third-party soil sampling: What you need to know*. Retrieved

February 6, 2025, from <https://www.cropnutrition.com/resource-library/third-party-soil-sampling-what-you-need-to-know/>

North Carolina State University Extension. (n.d.). *Careful soil sampling: The key to reliable*

soil test information. Retrieved February 6, 2025, from <https://content.ces.ncsu.edu/careful-soil-sampling-the-key-to-reliable-soil-test-information>

Soil Science Society of America. (n.d.). *Self-study exam: Soil sampling and testing*.

Retrieved February 6, 2025, from <https://www.soils.org/files/certifications/certified/education/self-study/exam-pdfs/243.pdf>

California Department of Transportation. (2011). *Trenching and shoring manual*.

Retrieved February 6, 2025, from <https://engineering.purdue.edu/~frosch/ftp/Talbott/11%20-%20References/files/California%20Trenching%20and%20Shoring%20Manual.pdf>

CB Insights. (n.d.). *Agriculture tech trends*. Retrieved February 6, 2025,

from <https://www.cbinsights.com/research/agriculture-tech-trends/>

Engineering News-Record. (n.d.). *Market data*. Retrieved February 6, 2025,

from https://www.enr.com/market_data

Gartner. (n.d.). *Market research methodologies*. Retrieved February 6, 2025,

from <https://www.gartner.com/en/research/methodologies/market-research>

Grand View Research. (n.d.). *Soil testing market size, share & trends analysis report*.

Retrieved February 6, 2025, from <https://www.grandviewresearch.com/industry-analysis/soil-testing-market>

Grand View Research. (n.d.). *Smart agriculture market size, share & trends analysis report*.

Retrieved February 6, 2025, from <https://www.grandviewresearch.com/industry-analysis/smart-agriculture-farming-market>

References

Following are used in the development of this FDR

International Food Policy Research Institute. (n.d.). *Publications*. Retrieved February 6, 2025, from <https://www.ifpri.org/publications>

Mordor Intelligence. (n.d.). *Geotechnical engineering market - growth, trends, COVID-19 impact, and forecasts*. Retrieved February 6, 2025, from <https://www.mordorintelligence.com/industry-reports/geotechnical-engineering-market>

NASA. (n.d.). *Earth science*. Retrieved February 6, 2025, from <https://science.nasa.gov/earth-science/> United Nations Environment Programme. (n.d.). *Resources*. Retrieved February 6, 2025, from <https://www.unep.org/resources>

United States Department of Agriculture, Economic Research Service. (n.d.). *Home page*. Retrieved February 6, 2025, from <https://www.ers.usda.gov>

The World Bank. (n.d.). *Agriculture and rural development*. Retrieved February 6, 2025, from <https://data.worldbank.org/topic/agriculture-and-rural-development>

Envirotecnicos. (2020). *AMS soil sampling*. Retrieved February 6, 2025, from <https://www.envirotecnicos.com/wp-content/uploads/2020/07/AMS-soil-sampling-envirotecnicos.pdf>